

Lab 2 — Task 6: Kubernetes Deployment Strategies and Scaling

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Objective

Using the lab2-app Deployment from Task 5, demonstrate a rolling update to a new image version, a failed update followed by a rollback, and scaling the number of replicas up and down.

Step 1 — Confirm the update strategy

The Deployment uses the RollingUpdate strategy with maxSurge: 1 and maxUnavailable: 0. With 3 replicas this means Kubernetes may create one extra pod (at most 4 exist during an update) and never takes a pod out of service before its replacement is ready, so the application stays available throughout an update.

```
kubectl get deployment lab2-app -o wide
kubectl describe deployment lab2-app | Select-String "StrategyType|RollingUpdateStrategy"
kubectl rollout history deployment/lab2-app
```

```
C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labssss\lab2-k8s-kit>curl.exe -s http://127.0.0.1:54924/healthcurl.exe -s http://127.0.0.1:54924/health
<!doctype html>
<html lang=en>
<title>404 Not Found</title>
<h1>Not Found</h1>
<p>The requested URL was not found on the server. If you entered the URL manually please check your spelling and try again.</p>
{"status":"ok"}

C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labssss\lab2-k8s-kit>kubectl get deployment lab2-app -o wide
NAME          READY   UP-TO-DATE   AVAILABLE   AGE    CONTAINERS   IMAGES               SELECTOR
lab2-app      3/3     3            3           12m    lab2-app     lab2-python-app:1.0  app=lab2-app

C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labssss\lab2-k8s-kit>kubectl describe deployment lab2-app | findstr "StrategyType RollingUpdateStrategy"
StrategyType:          RollingUpdate
RollingUpdateStrategy: 0 max unavailable, 1 max surge

C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labssss\lab2-k8s-kit>kubectl rollout history deployment/lab2-app
deployment.apps/lab2-app
REVISION  CHANGE-CAUSE
1         <none>

C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labssss\lab2-k8s-kit>
```

Figure 1 — Deployment strategy is RollingUpdate (0 max unavailable, 1 max surge); the history currently holds revision 1.

Step 2 — Build and load version 2.0 of the image

A small visible change is made to the application (the greeting message in app.py), and the image is rebuilt with a new tag and loaded into Minikube. A new tag is used instead of reusing 1.0 so that Kubernetes sees a change in the pod template and triggers an update.

```
docker build -t lab2-python-app:2.0 ./app
minikube image load lab2-python-app:2.0
minikube image ls
```

Step 3 — Perform a rolling update

In one terminal the pods are watched; in another the Deployment is updated to the 2.0 image. A change-cause annotation is added so the revision is described in the history.

```
# terminal 2
kubectl get pods -w

# terminal 1
kubectl set image deployment/lab2-app lab2-app=lab2-python-app:2.0
kubectl annotate deployment/lab2-app kubernetes.io/change-cause="Update to version 2.0"
kubectl rollout status deployment/lab2-app
kubectl rollout history deployment/lab2-app
```

```
C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labssss\lab2-k8s-kit>pip install flask
Requirement already satisfied: flask in c:\users\arsh\appdata\local\programs\python\python310\lib\site-packages (3.1.2)
Requirement already satisfied: blinker>=1.9.0 in c:\users\arsh\appdata\local\programs\python\python310\lib\site-packages (from flask) (1.9.0)
Requirement already satisfied: click>=8.1.3 in c:\users\arsh\appdata\local\programs\python\python310\lib\site-packages (from flask) (8.4.2)
Requirement already satisfied: itsdangerous>=2.2.0 in c:\users\arsh\appdata\local\programs\python\python310\lib\site-packages (from flask) (2.2.0)
Requirement already satisfied: jinja2>=3.1.2 in c:\users\arsh\appdata\local\programs\python\python310\lib\site-packages (from flask) (3.1.6)
Requirement already satisfied: markupsafe>=2.1.1 in c:\users\arsh\appdata\local\programs\python\python310\lib\site-packages (from flask) (3.0.3)
C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labssss\lab2-k8s-kit>cd C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labssss\lab2-k8s-kit
C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labssss\lab2-k8s-kit>kubectl set image deployment/lab2-app lab2-app=lab2-python-app:2.08.1.3->flask) (0.4.6)
deployment.apps/lab2-app image updated
C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labssss\lab2-k8s-kit>
C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labssss\lab2-k8s-kit>kubectl annotate deployment/lab2-app kubernetes.io/change-cause="Update to version 2.0" --overwrite
deployment.apps/lab2-app annotated

C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labssss\lab2-k8s-kit>kubectl rollout status deployment/lab2-app
Waiting for deployment "lab2-app" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "lab2-app" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "lab2-app" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "lab2-app" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "lab2-app" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "lab2-app" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "lab2-app" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "lab2-app" rollout to finish: 1 old replicas are pending termination...
deployment "lab2-app" successfully rolled out
```

Figure 2 — New pods are started one at a time and old pods are terminated only after their replacement is ready; the history now lists revision 2 ("Update to version 2.0").

Step 4 — Failed update and rollback

To show rollback, the Deployment is deliberately updated to an image tag that does not exist (99.0). The new pod cannot start (ErrImagePull / ImagePullBackOff), the rollout stalls, and — because maxUnavailable is 0 — the existing healthy pods keep serving traffic.

```
kubectl set image deployment/lab2-app lab2-app=lab2-python-app:99.0
kubectl rollout status deployment/lab2-app --timeout=30s
kubectl get pods
```

```
C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labsssss\lab2-k8s-kit>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
lab2-app-c95689f9c-f5p72            1/1     Running   0           2m5s
lab2-app-c95689f9c-hp62k            1/1     Running   0           119s
lab2-app-c95689f9c-sz74m            1/1     Running   0           113s

C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labsssss\lab2-k8s-kit>kubectl set image deployment/lab2-app lab2-app=lab2-python-app:99.0
deployment.apps/lab2-app image updated

C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labsssss\lab2-k8s-kit>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
lab2-app-7867fb8569-2kfft            0/1     ErrImagePull   0           4s
lab2-app-c95689f9c-f5p72            1/1     Running        0          2m40s
lab2-app-c95689f9c-hp62k            1/1     Running        0          2m34s
lab2-app-c95689f9c-sz74m            1/1     Running        0          2m28s
```

Figure 3 — The bad update is stuck: the new pod is in ImagePullBackOff while the previous pods remain Running and Ready.

The failed rollout is then reverted with rollout undo, which returns the Deployment to the previous revision. The image in use and the history are checked afterwards.

```
kubectl rollout undo deployment/lab2-app
kubectl rollout status deployment/lab2-app
kubectl rollout history deployment/lab2-app
kubectl get deployment lab2-app -o jsonpath="{.spec.template.spec.containers[0].image}"
kubectl get pods
```

```
C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labsssss\lab2-k8s-kit>kubectl set image deployment/lab2-app lab2-app=lab2-python-app:99.0

C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labsssss\lab2-k8s-kit>kubectl rollout undo deployment/lab2-app
deployment.apps/lab2-app rolled back

C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labsssss\lab2-k8s-kit>kubectl rollout status deployment/lab2-app
deployment "lab2-app" successfully rolled out

C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labsssss\lab2-k8s-kit>kubectl rollout history deployment/lab2-app
deployment.apps/lab2-app
REVISION  CHANGE-CAUSE
1         <none>
3         Update to version 2.0
4         Update to version 2.0

C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labsssss\lab2-k8s-kit>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
lab2-app-c95689f9c-f5p72            1/1     Running   0           4m44s
lab2-app-c95689f9c-hp62k            1/1     Running   0           4m38s
lab2-app-c95689f9c-sz74m            1/1     Running   0           4m32s
```

Figure 4 — rollout undo restores the last good revision (image lab2-python-app:2.0); all pods return to Running 1/1.

Step 5 — Scale the replicas

The application is scaled up to 5 replicas, then down to 2, and finally back to the original 3. Scaling changes only the desired replica count; the ReplicaSet creates or removes pods to match it.

```
kubectl scale deployment lab2-app --replicas=5
kubectl get deployment lab2-app
kubectl get pods

kubectl scale deployment lab2-app --replicas=2
kubectl get pods

kubectl scale deployment lab2-app --replicas=3
```

```
C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labssss\lab2-k8s-kit>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
lab2-app-c95689f9c-f5p72            1/1     Running   0           4m44s
lab2-app-c95689f9c-hp62k            1/1     Running   0           4m38s
lab2-app-c95689f9c-sz74m            1/1     Running   0           4m32s

C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labssss\lab2-k8s-kit>kubectl scale deployment lab2-app --replicas=5
deployment.apps/lab2-app scaled

C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labssss\lab2-k8s-kit>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
lab2-app-c95689f9c-f5p72            1/1     Running   0           5m50s
lab2-app-c95689f9c-hp62k            1/1     Running   0           5m44s
lab2-app-c95689f9c-rr6s9            0/1     Running   0           4s
lab2-app-c95689f9c-sz74m            1/1     Running   0           5m38s
lab2-app-c95689f9c-xz1tn            0/1     Running   0           4s

C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labssss\lab2-k8s-kit>kubectl scale deployment lab2-app --replicas=2
deployment.apps/lab2-app scaled

C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labssss\lab2-k8s-kit>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
lab2-app-c95689f9c-f5p72            1/1     Terminating   0           6m
lab2-app-c95689f9c-hp62k            1/1     Running         0           5m54s
lab2-app-c95689f9c-rr6s9            1/1     Terminating   0           14s
lab2-app-c95689f9c-sz74m            1/1     Running         0           5m48s
lab2-app-c95689f9c-xz1tn            1/1     Terminating   0           14s

C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labssss\lab2-k8s-kit>kubectl scale deployment lab2-app --replicas=3
deployment.apps/lab2-app scaled
```

Figure 5 — Scaling to 5 adds two pods (5/5 ready); scaling to 2 terminates three pods; the Deployment is finally set back to 3 replicas.

Step 6 — Check traffic distribution across replicas

With the tunnel from Task 5 open (minikube service lab2-app-svc --url in a separate terminal), several requests are sent through the Service. Each response contains the hostname of the pod that handled it, so different hostnames show that the Service spreads requests across the replicas.

```
1..6 | ForEach-Object { curl.exe -s http://127.0.0.1:<port>/ }
```

```
C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labssss\lab2-k8s-kit>kubectl scale deployment lab2-app --replicas=3
deployment.apps/lab2-app scaled

C:\Users\ARSH\OneDrive\Desktop\PDFs\zeee\labssss\lab2-k8s-kit>curl.exe -s http://127.0.0.1:54924/
{"data_dir":"/data","hostname":"lab2-app-c95689f9c-sz74m","message":"Hello from the Lab 2 containerized Python ap
p!","time_utc":"2026-09-30T09:38:33.893362+00:00","visits_recorded_in_this_container":1}
```

Figure 6 — Responses come from different pods (different hostname values), confirming load distribution by the Service.

Summary of observations

Operation	Command	Behavior observed
Rolling update	<code>kubectl set image ...:2.0</code>	Pods replaced gradually; never fewer than 3 ready pods; new revision recorded.
Failed update	<code>kubectl set image ...:99.0</code>	New pod stuck in ImagePullBackOff; old pods kept serving traffic.
Rollback	<code>kubectl rollout undo</code>	Deployment returned to the last good revision; all pods Running.
Scale up	<code>kubectl scale --replicas=5</code>	Two extra pods created; Deployment 5/5 ready.
Scale down	<code>kubectl scale --replicas=2</code>	Surplus pods terminated; Deployment 2/2 ready.

Result

The Deployment was updated from version 1.0 to 2.0 with a zero-downtime rolling update, a deliberately broken update was contained by the `maxUnavailable: 0` setting and reverted with `kubectl rollout undo`, and the application was scaled between 2 and 5 replicas, with requests distributed across the running pods. Task 6 is complete.